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Feasibility Recovery for Fixed-Horizon Multi-UAV Cooperative Flight Trajectory Optimization in High-Conflict Regions

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Abstract

Cooperative flight through intersecting routes and narrow passages requires UAVs to avoid obstacles and one another while reaching prescribed endpoints within a common fixed horizon. Local collision-avoidance corridors can conflict with motion limits and prevent a direct optimization from producing an executable plan. This paper develops a layered optimization framework that addresses these conflicts without extending the flight horizon. A hard-corridor formulation provides a fast initial path for compatible tasks. When it fails to yield an accepted trajectory, separate penalties on kinematic inconsistency and inter-UAV corridor violations allow intermediate trajectories to reshape the corridors. A second recovery layer responds to violations in the reconstructed motion by tightening consistency requirements and solving with exact kinematics. Endpoint control correction and exact integration produce rest-to-rest motions, which undergo complete-interval clearance and motion-limit checks. The local subproblems retain a sparse convex quadratic structure. Simulations examine centralized teams of up to 20 UAVs and include closed-loop rigid-body tracking under disturbances. In the examined winding tasks, the recovery layers obtain executable motions after the initial hard-corridor formulation fails, while the direct path handles compatible cases efficiently.

Keywords: relative safe corridor; lightweight iterative optimization method (LIOM); trajectory planning; cooperative trajectory planning; computational optimal control

1. Introduction

Cooperative trajectory planning enables multiple unmanned aerial vehicles (UAVs) to execute coordinated missions in a shared three-dimensional workspace. When several vehicles approach an intersection or a narrow passage, their trajectories must satisfy obstacle clearance, mutual separation, and motion limits simultaneously. Aerial-swarm research has established the importance of these coupled requirements [1], while agile quadrotor teams and coordinated formation transitions demonstrate their practical relevance [2,3]. Recent work combines three-dimensional reciprocal velocity obstacles with multi-agent reinforcement learning to coordinate collision avoidance within structured urban airspace [4].

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Computational efficiency is essential when a feasible coordinated motion must be obtained within a limited planning interval. Parallel stitching addresses this requirement in

online parking replanning by evaluating candidate connecting trajectories in parallel and preserving continuity with the ongoing motion [5]. Enhanced sequential convex programming has been integrated with model predictive control to address formation tracking and correct linearization errors that can compromise feasibility [6]. Conflict-free multi-robot planning coupled with predictive control connects coordination decisions to feasible motion [7]. Cooperative exploration combines dynamic Voronoi partitioning with learned obstacle avoidance [8], while multi-UAV coverage planning jointly considers task allocation and travel time under repeated round-trip operation [9]. Together, these studies motivate methods that account for coupled coordination, safety, and timing requirements while keeping computation manageable.

High-conflict regions arise where assigned routes intersect or several UAVs compete for a shared passage around obstacles. Safe corridors express collision avoidance through conservative linear restrictions. A useful reference can nevertheless produce local corridors that conflict with one another or with bounded motion. Under a prescribed completion time, all of these requirements must be reconciled within the original flight horizon. A failed local solve therefore calls for a mechanism that can revise the coordinated motion and its surrounding geometry while preserving the task boundaries.

This paper develops a lightweight iterative optimization method (LIOM) for fixed-horizon, rest-to-rest cooperative flight. The planner first attempts a hard relative-corridor QP. An accepted result is returned immediately. Otherwise, a relaxed formulation supplies intermediate candidates for corridor reconstruction. A second recovery layer handles candidates whose reconstructed motion violates a physical limit, tightening the kinematic tolerance and attempting exact finishing. The double-integrator model yields affine defects, so both stages remain convex QPs with linear constraints. The recovery loop treats kinematic defects and relative-corridor slack separately. Exact control integration and complete-interval checking connect the numerical candidates to the prescribed physical task.

The main contributions are twofold:

1. A fixed-horizon cooperative formulation combines reachable-speed corridor padding with an affine two-penalty QP. The deadline and rest-to-rest endpoints remain fixed throughout refinement. An explicit feasible-point construction establishes nonemptiness of the relaxed local problem, while conservative polyhedral motion bounds and box-distance screening retain sparse linear constraints.
2. An acceptance-driven solution procedure combines a hard-QP entry, capped continuation, corridor reconstruction, and endpoint moment correction. Failed continuous-motion checks trigger tighter recovery and an exact-kinematics finish in the current corridors. Complete-interval checks on the integrated output determine acceptance. This procedure concentrates additional recovery effort on attempts that require it.

The evaluation follows the three computational paths: direct hard-QP solution, relaxed corridor recovery, and acceptance-driven finishing. Matched hard formulations receive the same reconstruction opportunities. Independently checked winding tasks expose local corridor conflicts, while scaling, parameter, and tracking studies characterize computation and execution.

2. Related Work

2.1. Coordinated Paths, Prediction, and Safety Control

Cooperative planning must establish a compatible passing pattern and motions that vehicles can execute. Preiss et al. [3] combined discrete coordination with polynomial refinement and downwash-aware separation. Hönig et al. [10] connected collision-annotated roadmaps to continuous quadrotor trajectories, while CAPT [11] jointly selected

assignments and trajectories for interchangeable robots. Discrete coordination can resolve the order of passage before continuous refinement. When assignments and arrival times are prescribed, the refinement must also respect the motion budget associated with each fixed endpoint. Changing an assignment or allocating more time is then outside the planning decision.

Distributed model predictive control (MPC) coordinates vehicles through predicted neighbor motions. Luis and Schoellig [12] introduced on-demand collision constraints and penalized relaxation; Luis et al. [13] developed online trajectory generation with continuous input parameterization and event-triggered replanning. Buffered Voronoi cells [14] allocate local free space through relative-position geometry. Communication-efficient predictive formation control [15] reduces information exchange using autoencoders. These methods address local interaction and online update costs. Their rolling prediction windows differ from a prescribed completion horizon: the latter ties every remaining control interval to terminal position and velocity conditions that must be met at the same fixed time.

Control barrier functions (CBFs) constrain the current control action. Gonçalves et al. [16] constructed safe drone acceleration fields with a quadratic program (QP), followed by thrust and angular-velocity tracking. Mestres et al. [17] distributed coupled safety constraints through auxiliary mismatch variables. These methods provide feasibility or safety results under their stated assumptions. Their control objectives concern instantaneous actions; a complete rest-to-rest motion with a fixed deadline requires additional temporal coordination. Their QP-based control updates provide a complementary level of coordination.

Deadlock studies further distinguish local safety from sustained progress. Grover et al. [18] characterized reactive-controller deadlock through optimality conditions. Jankovic et al. [19] linked gridlock propensity to equilibrium stability and developed predictor–corrector coordination. Chandra et al. [20] incorporated deadlock prevention into decentralized MPC with discrete-time CBFs. These results motivate explicit treatment of conflicting avoidance decisions. They also show why success under a fixed arrival time is a stronger task requirement than the availability of a safe instantaneous action or eventual progress toward a goal.

2.2. Corridor-Based and Centralized Optimization

Safe corridors turn geometric avoidance into conservative local constraints. Park and Kim [21] used relative safe flight corridors with relaxation and reciprocal avoidance for online quadrotor planning in dynamic environments. Şenbaşlar et al. [22] developed RLSS using linear spatial separations and polynomial QPs, with duration adjustment when derivative limits require it. Hard corridors are effective when their intersection with the motion constraints is nonempty. When that intersection is empty, a solver must be complemented by a change in local geometry, relaxation, or timing. Fixed-time tasks retain the first two options while excluding horizon extension.

Online planners address additional execution and communication requirements. EGO-Swarm [23] combines local optimization with perception and topological alternatives. MADER [24] uses asynchronous trajectory checking and commitment; Robust MADER [25] accounts for communication delay. PRIMER [26] learns perception-aware position–yaw planning while maintaining deconfliction. Park et al. [27] coordinate waypoints and subgoals with convergence and deadlock-prevention results under specified communication conditions. These capabilities are valuable for distributed operation. Centralized fixed-horizon planning instead couples the full joint motion in a known static map before execution.

Centralized formulations expose the full coupling among vehicles. Adaptive-scaling constrained optimization (ASCO) [28] changes the active collision-risk range during nonlinear cooperative planning. Priority-sharing intersection management [29] combines social-force references, optimization, and reservations. Sequential convex programming (SCP) [30] repeatedly updates local affine collision restrictions. Each approach uses a different response to difficult intermediate problems: active-set adjustment, coordination restructuring, or local approximation. In the fixed-team, fixed-endpoint problem considered here, intermediate failures must be handled while preserving the assigned task. A hard iterative corridor comparison is therefore especially informative: it tests whether reconstruction alone can provide the needed adjustment.

2.3. From Iterative Corridors to Accepted Fixed-Time Motion

Li et al. [31] introduced lightweight iterative optimization for parking, combining box corridors, softened nonlinear equalities, and repeated reconstruction. Bu et al. [32] extended cooperative corridor planning with spatial and relative boxes, relative-corridor slack, kinematic penalties, and social-force initialization. These works establish the iterative corridor and penalty principles used here. The present formulation fixes each UAV's terminal position and zero terminal velocity at a common prescribed time. Its affine integration defects yield convex QPs, and reachable-speed padding connects nodal clearance to inter-node motion. A closed-form control-moment correction then enforces both terminal conditions before the continuous trajectory is checked.

The solution procedure gives a hard formulation a direct execution path whenever its returned motion passes the common checks. Unsuccessful entry or physical acceptance invokes relaxed reconstruction, tolerance adjustment, and exact finishing. These branches address distinct failures: unavailable local hard motions, residual kinematic mismatch, and changes introduced by output reconstruction. Matched hard iterative methods and independently checked conflict cases examine the role of each computational path. Table 1 summarizes the methodological boundaries. For execution, linking planning to a tracking model is equally relevant; Huang et al. [33] demonstrated this connection through smooth vehicle paths and control-oriented cosimulation. Here it is examined with a multirotor rigid-body simulation.

3. Problem Formulation

3.1. Fixed-Horizon Rest-to-Rest Task

Consider N labeled UAVs in a known workspace $\mathcal{W} = [\underline{\mathbf{w}}, \overline{\mathbf{w}}] \subset \mathbb{R}^3$. Static obstacles are closed axis-aligned bounding boxes (AABBs), $\mathcal{O}_b = [\underline{\mathbf{o}}_b, \overline{\mathbf{o}}_b]$, $b = 1, \ldots, B$. Each UAV is represented by a sphere of radius r, with a required surface clearance c. Task identities are fixed. UAV i moves from its assigned start $\mathbf{s}_i$ to goal $\mathbf{g}_i$ over the prescribed common flight duration $T_f > 0$. This duration is fixed before planning and shared by every UAV; it is distinct from computation time. The boundary conditions are

$$\mathbf{p}_i(0) = \mathbf{s}_i, \quad \mathbf{p}_i(T_f) = \mathbf{g}_i, \quad \mathbf{v}_i(0) = \mathbf{v}_i(T_f) = \mathbf{0}. \tag{1}$$

Both endpoints are inside the free workspace and mutually separated. In the following, $\mathbf{v}_i^0 = \mathbf{v}_i^f = \mathbf{0}$ denote these prescribed endpoint velocities. Vector inequalities are understood componentwise.

3.2. Motion Model and Reduced Variables

We consider hovering-capable rotary-wing UAVs. Their planning-level translational dynamics are

$$\dot{\mathbf{p}}_i = \mathbf{v}_i, \qquad \dot{\mathbf{v}}_i = \mathbf{a}_i. \tag{2}$$

Table 1. Representative formulations and their response to difficult local planning problems. The common completion deadline differs from a rolling prediction horizon.

Method	Model and coordination	Timing / boundary commitment	Recovery or acceptance mechanism
Li et al. [31]	Nonlinear parking model; spatial boxes	Optimized parking duration	Softened nonlinear equalities and iterative corridor reconstruction
Bu et al. [32]	Cooperative vehicle model; STC/RSTC	Fixed traverse horizon; exit-region endpoints	Kinematic and relative-corridor penalties; repeated construction
Park and Kim [21]	Online multi-quadrotor relative corridors	Receding-horizon motion	Relative-corridor relaxation and reciprocal avoidance
RLSS [22]	Decentralized polynomial QPs	Adjustable segment durations	Linear separation and temporal rescaling for derivative limits
ASCO [28]	Centralized nonlinear cooperative planning	Optimized terminal time	Adaptation of the active collision-risk range
CBF control [16,19]	State-dependent safety-control QPs	Instantaneous control and navigation	Feasible safety filters and deadlock-aware policies under stated assumptions
SCP [30]	Centralized convex trajectory approximations	Prescribed trajectory duration	Updated affine separation constraints
Proposed LIOM	Affine multi-UAV QPs; spatial and relative regions	Fixed common time, positions, and rest endpoints	Hard entry, relaxed reconstruction, endpoint moment correction, and complete-interval acceptance

Let $h = T_f/K$ and $t_k = kh$, $k = 0, \ldots, K$, and take acceleration to be constant on each interval. Exact integration yields

$$\mathbf{v}_{i,k+1} = \mathbf{v}_{i,k} + h\mathbf{a}_{i,k}, \tag{3a}$$

$$\mathbf{p}_{i,k+1} = \mathbf{p}_{i,k} + \frac{h}{2}(\mathbf{v}_{i,k} + \mathbf{v}_{i,k+1}). \tag{3b}$$

Eliminating acceleration reduces the physical decision fields to $\mathbf{P}, \mathbf{V}$:

$$\mathbf{a}_{i,k} = \frac{\mathbf{v}_{i,k+1} - \mathbf{v}_{i,k}}{h}, \qquad \mathbf{E}_{i,k} = \mathbf{p}_{i,k+1} - \mathbf{p}_{i,k} - \frac{h}{2}(\mathbf{v}_{i,k+1} + \mathbf{v}_{i,k}). \tag{4}$$

An exact discrete trajectory has $\mathbf{E}_{i,k} = \mathbf{0}$. During optimization, this relation is relaxed to allow the position and velocity fields to adjust jointly. Section 6.3 gives the final control-integration procedure.

The Euclidean speed and acceleration limits are $v_{\max}$ and $a_{\max}$. The planning model uses an isotropic translational envelope. Attitude-dependent actuation enters the rigid-body tracking model in Section 7.8. To retain linear constraints while reducing the directional conservatism of an inscribed cube, define

$$\mathcal{P} = \operatorname{conv}\Big(\{\pm\mathbf{e}_1, \pm\mathbf{e}_2, \pm\mathbf{e}_3\} \cup \{(\pm 1, \pm 1, \pm 1)/\sqrt{3}\}\Big) = \{\mathbf{x} : U\mathbf{x} \le \mathbf{b}\}. \tag{5}$$

Every vertex has unit Euclidean norm. Convexity of the unit ball therefore gives $\mathcal{P} \subseteq \{\mathbf{x} : \|\mathbf{x}\|_2 \le 1\}$. This 14-vertex, 24-face polytope contains the inscribed Cartesian cube and attains the full limit along each coordinate axis. The motion restrictions are

$$U\mathbf{v}_{i,k} \le v_{\max}\mathbf{b}, \qquad U(\mathbf{v}_{i,k+1} - \mathbf{v}_{i,k})/h \le a_{\max}\mathbf{b}. \tag{6}$$

These are sufficient inner approximations of the Euclidean limits. Final acceptance tests the Euclidean speed and acceleration of the reconstructed motion.

3.3. Workspace and Collision Conditions

For $\mathbf{p} \in \mathbb{R}^3$, define the component gap to an obstacle box as

$$\mathbf{d}_b(\mathbf{p}) = \max\{\underline{\mathbf{o}}_b - \mathbf{p},\ \mathbf{p} - \overline{\mathbf{o}}_b,\ \mathbf{0}\}. \tag{7}$$

The physical requirements, imposed over the complete horizon, are

$$\underline{\mathbf{w}} + r\mathbf{1} \le \mathbf{p}_i(t) \le \overline{\mathbf{w}} - r\mathbf{1}, \tag{8a}$$

$$\|\mathbf{d}_b(\mathbf{p}_i(t))\|_2 \ge r + c, \qquad \forall b, \tag{8b}$$

$$\|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2 \ge 2r + c, \qquad \forall i < j. \tag{8c}$$

Here $t \in [0, T_f]$ is physical flight time, $\mathbf{p}_i(t)$ is the center position of UAV i, and c is the required surface-to-surface clearance. The indices b and $i < j$ cover all obstacles and UAV pairs. These conditions apply to the entire motion of the spherical planning footprint.

3.4. Acceleration-Effort Objective

With task completion fixed by (1), the nominal objective is

$$C(\mathbf{V}) = w_a h \sum_{i=1}^{N} \sum_{k=0}^{K-1} \left\| \frac{\mathbf{v}_{i,k+1} - \mathbf{v}_{i,k}}{h} \right\|_2^2, \qquad w_a > 0. \tag{9}$$

The coefficient w_a weights acceleration effort, and $C(\mathbf{V})$ is the weighted sum of squared accelerations integrated over time. This objective promotes smooth velocity changes in the recovered motion. For a dimensionless objective in SI coordinates, w_a has units $\mathrm{s}^3/\mathrm{m}^2$. The flight duration T_f is prescribed independently of computation time. The outer procedure prioritizes an accepted motion within its computation budget. The quadratic objective guides each local solve.

The nominal problem minimizes (9) subject to exact dynamics, fixed boundary states, motion limits, and continuous collision avoidance. Geometric avoidance makes this problem nonconvex. Spatial and relative corridors provide conservative local linear restrictions, and the recovery mechanism handles their compatibility with the motion model.

4. Three-Dimensional Corridor-Based Collision Avoidance

4.1. From a Reference Motion to Linear Restrictions

Assume synchronized reference nodes $\widehat{\mathbf{p}}_{i,k}$ are available. Their generation is described in Section 5. The reference determines local boxes that provide conservative linear restrictions for collision avoidance. Each box has three lower and three upper coordinate bounds. Independent six-face expansion produces spatial safe travel corridors (STCs) and relative safe travel corridors (RSTCs) indexed by time.

4.2. Spatial Safe Travel Corridors

To account for movement between nodes, the construction includes temporal padding. For an exactly consistent motion satisfying the Euclidean limits, a reachable-speed bound at node k is

$$b_{i,k} = \min\{v_{\max},\ a_{\max} t_k,\ a_{\max}(T_f - t_k)\}. \tag{10}$$

The two acceleration terms bound the speed reachable from rest and the speed that can still be brought to rest by T_f. Thus $b_{i,k}$ tightens padding near either endpoint. Every time

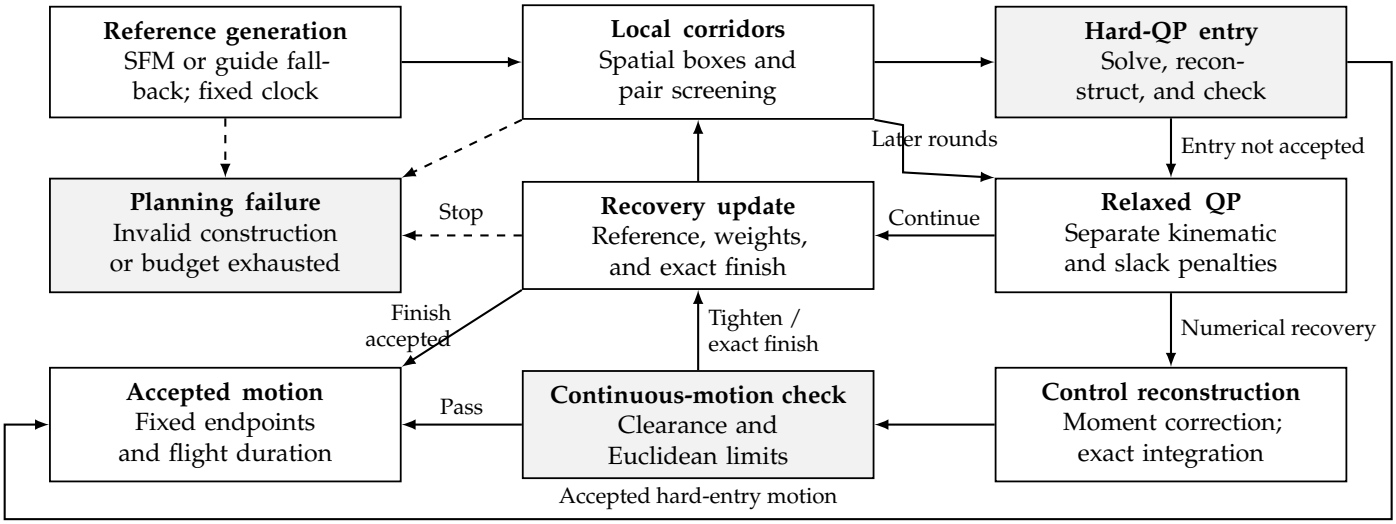


Figure 1. Fixed-horizon cooperative trajectory optimization. The hard entry includes reconstruction and physical checking. Relaxed candidates enter the recovery loop; exact finishing provides a further acceptance branch. All returned motions pass complete-interval checks. Dashed links mark exits to planning failure.

in the horizon is at most $h/2$ from a node. With numerical guard ϵ_g, the corresponding padding is

$$q_{i,k} = \min\left\{ \frac{v_{\max} h}{2},\ \frac{b_{i,k} h}{2} + \frac{a_{\max} h^2}{8} \right\} + \epsilon_g. \tag{11}$$

Obstacles are inflated componentwise and the workspace is inset:

$$\mathcal{O}_b^{i,k} = [\underline{\mathbf{o}}_b - (r + c + q_{i,k})\mathbf{1}, \overline{\mathbf{o}}_b + (r + c + q_{i,k})\mathbf{1}], \tag{12a}$$

$$\mathcal{W}_{i,k} = [\underline{\mathbf{w}} + (r + q_{i,k})\mathbf{1}, \overline{\mathbf{w}} - (r + q_{i,k})\mathbf{1}]. \tag{12b}$$

Axiswise inflation conservatively covers Euclidean obstacle dilation. The STC is a box

$$\Gamma_{i,k} = [\underline{\gamma}_{i,k}, \overline{\gamma}_{i,k}] \subset \mathcal{W}_{i,k}, \qquad \Gamma_{i,k} \cap \mathcal{O}_b^{i,k} = \varnothing \quad \forall b. \tag{13}$$

Starting from a valid seed, the algorithm expands the lower and upper faces along x, y, and z in turn, following incremental local-box growth [31]. A step is accepted only if the box remains inside the inset workspace and outside every inflated obstacle. Each face advances by an increment Δ_c, up to a prescribed expansion limit ℓ_s. When a proposed step intersects an inflated obstacle or leaves the inset workspace, bisection locates the last admissible face position. That face is frozen while the remaining faces continue to grow. Expansion ends when every face is blocked or has reached its limit. Figure 2(a) illustrates the seed, an intermediate box, and the final corridor.

A reference node can be invalid after temporal inflation even when its uninflated position is free. Such a node is first clipped to the inset workspace. If it remains inside an inflated obstacle, the existing static guide is searched for the nearest admissible projected point. If no guide candidate is available, admissible projections beyond obstacle faces are tested. An admissible seed is required to construct the local corridor.

The resulting static-avoidance constraint is simply

$$\underline{\gamma}_{i,k} \le \mathbf{p}_{i,k} \le \overline{\gamma}_{i,k}. \tag{14}$$

The spatial box is retained as a hard constraint. Its padding accounts for inter-node motion under the stated speed and acceleration limits.

4.3. Relative Safe Travel Corridors

For synchronized UAVs i and j, introduce

$$\mathbf{r}_{ij,k} = \mathbf{p}_{i,k} - \mathbf{p}_{j,k} \in \mathbb{R}^3. \tag{15}$$

Collision corresponds to this relative vector entering a sphere around the origin. A relative reachable-speed bound and protected radius are

$$b_{ij,k} = \min\{2v_{\max},\, 2a_{\max}t_k,\, 2a_{\max}(T_f - t_k)\}, \tag{16a}$$

$$R_{ij,k} = 2r + c + \min\left\{v_{\max}h,\, \frac{b_{ij,k}h}{2} + \frac{a_{\max}h^2}{4}\right\} + 2\epsilon_g. \tag{16b}$$

Inter-node displacement bound.

Consider a kinematically consistent motion satisfying the speed and acceleration limits, and let t_k be a nearest node to t, with $\tau = |t - t_k| \le h/2$. Integrating the velocity bound, or integrating from the node using $\|\mathbf{v}_i(t_k)\|_2 \le b_{i,k}$ and the acceleration bound, gives

$$\left\|\mathbf{p}_i(t) - \mathbf{p}_{i,k}\right\|_2 \le \min\{v_{\max}\tau,\; b_{i,k}\tau + \tfrac{1}{2}a_{\max}\tau^2\} \le q_{i,k} - \epsilon_g. \tag{17}$$

The same argument for $\mathbf{r}_{ij}(t) = \mathbf{p}_i(t) - \mathbf{p}_j(t)$ uses relative speed and acceleration bounds $2v_{\max}$ and $2a_{\max}$:

$$\left\|\mathbf{r}_{ij}(t) - \mathbf{r}_{ij,k}\right\|_2 \le \min\{2v_{\max}\tau,\; b_{ij,k}\tau + a_{\max}\tau^2\}. \tag{18}$$

Substitution of $\tau \le h/2$ yields the $h^2/8$ and $h^2/4$ terms in (11) and (16). By the triangle inequality, nodal clearance enlarged by the corresponding padding is sufficient to preserve the required continuous-time clearance over the neighboring half intervals. This conclusion applies to kinematically consistent motion satisfying the stated limits. Relaxed candidates pass through the reconstruction and acceptance stages in Section 6.

Relative boxes are unnecessary when the corresponding STCs already separate all possible positions. Define

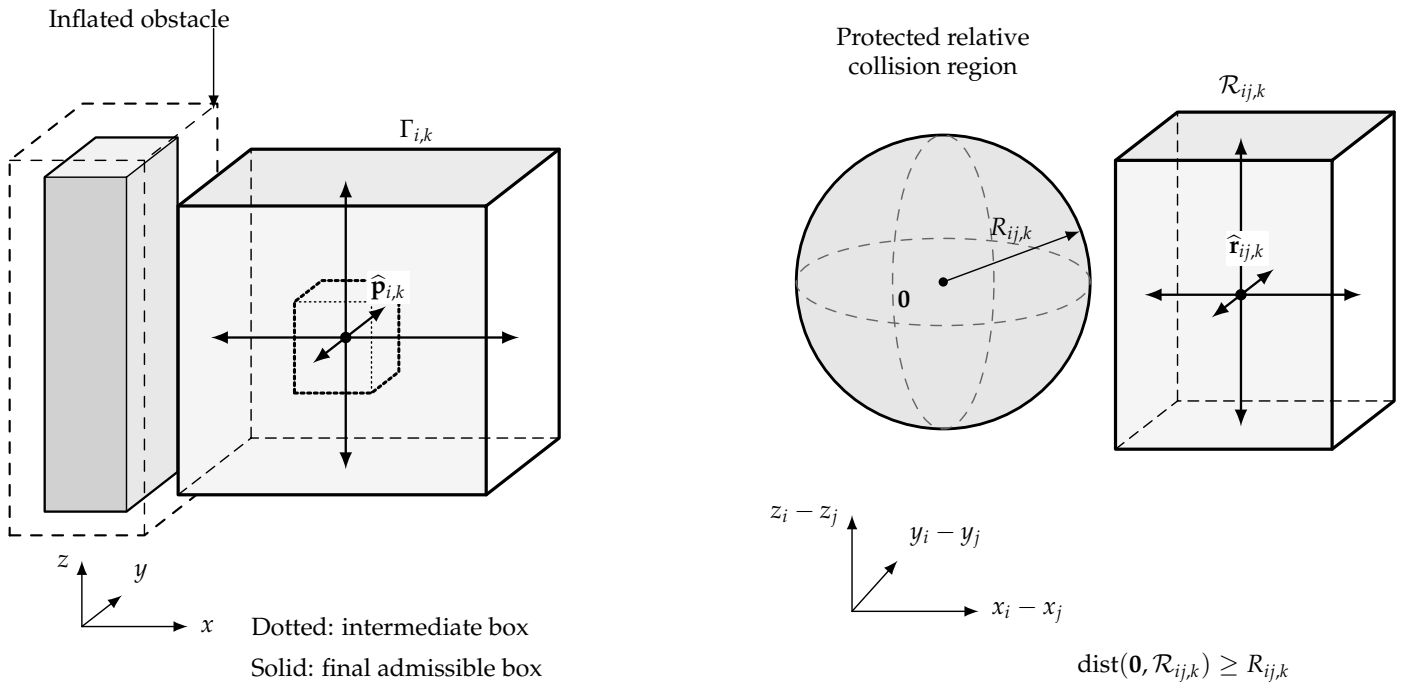
$$\mathbf{g}_{ij,k} = \max\{\underline{\gamma}_{i,k} - \overline{\gamma}_{j,k},\, \underline{\gamma}_{j,k} - \overline{\gamma}_{i,k},\, \mathbf{0}\}. \tag{19}$$

Only pair–node triples with $\left\|\mathbf{g}_{ij,k}\right\|_2 < R_{ij,k}$ belong to the active set $\mathcal{A}$. The minimum distance between the two spatial boxes is exactly $\left\|\mathbf{g}_{ij,k}\right\|_2$. A discarded pair cannot violate the protected nodal distance anywhere within those boxes. The same screening rule is used for hard-box and hard-plane comparisons. Every pair remains in final continuous-interval validation.

For each active triple, a box $\mathcal{R}_{ij,k} = [\underline{\boldsymbol{\eta}}_{ij,k}, \overline{\boldsymbol{\eta}}_{ij,k}]$ is grown around the reference relative position, subject to

$$\left\|\max\{\underline{\boldsymbol{\eta}}_{ij,k}, -\overline{\boldsymbol{\eta}}_{ij,k}, \mathbf{0}\}\right\|_2 \ge R_{ij,k}. \tag{20}$$

The left side is the exact minimum Euclidean distance from the origin to the box. Thus every point in the unrelaxed relative box lies outside the protected sphere. The relative box grows through the same six-face procedure, with step Δ_c and expansion limit ℓ_r. Each proposed face movement is accepted only while (20) holds. A seed inside the protected sphere is first displaced radially beyond its boundary by a small guard distance. For coincident seeds, the direction is obtained from the assigned start–goal geometry. Figure 2(b) shows the construction in relative-position space.



(a) Spatial corridor growth

(b) Relative corridor growth

Figure 2. Six-face corridor growth. (a) A spatial box expands from a reference seed; each face stops at an inflated obstacle, the inset workspace, or its expansion limit. (b) A relative box expands while remaining outside the protected sphere. The arrows denote independent growth along the positive and negative coordinate directions.

4.4. Slackened Relative Bounds

Independent spatial and relative boxes can conflict. One nonnegative scalar $\delta_{ij,k}$ therefore relaxes all three relative coordinates of each active triple:

$$\underline{\boldsymbol{\eta}}_{ij,k} - \delta_{ij,k}\mathbf{1} \leq \mathbf{p}_{i,k} - \mathbf{p}_{j,k} \leq \overline{\boldsymbol{\eta}}_{ij,k} + \delta_{ij,k}\mathbf{1}, \qquad \delta_{ij,k} \geq 0. \tag{21}$$

The scalar slack, measured in length units, quantifies compatibility with the selected relative box. A positive value can occur even when the UAVs remain physically separated, since the box is conservative. At zero slack, active pairs satisfy the protected nodal-distance condition through (20), and screened pairs are separated by their spatial boxes. The final integrated motion is checked against the original physical clearance conditions.

5. Synchronized Reference Trajectory Generation

A coordinated reference determines the initial corridors and supplies a warm start for optimization. Static A* guidance [34] provides routes through the environment, and a synchronous social force model (SFM) [35] establishes an initial passing pattern. Parameterized social-force references can guide the selection of cooperative maneuvers [29]. The same scalar timing rule places every geometric path on the prescribed rest-to-rest horizon.

5.1. Geometric Guidance and Cooperative Interaction

For each UAV, a collision-free start–goal segment is used when available. Otherwise, a three-dimensional A* search in the inflated obstacle map generates a guide path, which is shortened by line-of-sight checks. The inflation includes the UAV footprint, clearance, and a margin for inter-node motion.

Let $\mathbf{q}_i$ be a point a look-ahead distance ℓ_g along the guide, and let $d_i = \|\mathbf{g}_i - \mathbf{p}_i\|_2$ be the distance to the goal. A preferred speed v_{ref}, goal-approach time scale τ_g, and velocity-response time scale τ_v define

$$\mathbf{v}_i^{\text{pref}} = \min\{v_{\text{ref}}, d_i/\tau_g\}\frac{\mathbf{q}_i - \mathbf{p}_i}{\max\{\|\mathbf{q}_i - \mathbf{p}_i\|_2, \ell_g\}}, \qquad \mathbf{f}_i^{\text{att}} = \frac{\mathbf{v}_i^{\text{pref}} - \mathbf{v}_i}{\tau_v}. \tag{22}$$

The guide point advances monotonically along the route, and the preferred speed decreases near the goal.

For neighboring UAVs, define $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ and $\widehat{\mathbf{r}}_{ij} = (\mathbf{p}_i - \mathbf{p}_j)/d_{ij}$. For $\epsilon_d < d_{ij} < d_{\text{cut}}$, the pairwise response is

$$\mathbf{f}_{ij} = A_p \exp\left[\min\left\{\chi_{\max}, \frac{2r + c + b_p - d_{ij}}{B_p}\right\}\right](\widehat{\mathbf{r}}_{ij} + \kappa\,\mathbf{e}_z \times \widehat{\mathbf{r}}_{ij}). \tag{23}$$

Here A_p sets the repulsion strength, B_p its decay length, b_p an interaction buffer, and $\chi_{\max}$ a numerical cap. Outside this distance range the pair response is zero. In particular, the small guard ϵ_d suppresses normalization at coincident positions. The coefficient κ controls the tangential component, which encourages a consistent passing direction around the common vertical axis $\mathbf{e}_z$. The paired contributions are equal and opposite before goal tapering; their directions depend on geometry.

For compact notation, set $\psi(d; s) = \exp[\min\{\chi_{\max}, (s - d)/B_p\}]$. The nearest point on obstacle b is $\mathbf{c}_{ib} = \text{clip}(\mathbf{p}_i, \underline{\mathbf{o}}_b, \overline{\mathbf{o}}_b)$, with $d_{ib} = \|\mathbf{p}_i - \mathbf{c}_{ib}\|_2$. The obstacle and wall responses are

$$\mathbf{f}_i^{\text{obs}} = A_o \sum_{b:\,\epsilon_d < d_{ib} < d_o} \psi(d_{ib}; s_o)\frac{\mathbf{p}_i - \mathbf{c}_{ib}}{d_{ib}}, \qquad s_o = r + c + v_{\max}h/2 + \epsilon_g, \tag{24a}$$

$$f_{i,d}^{\text{wall}} = A_o\left[\psi(\ell_{i,d}^-; b_w) - \psi(\ell_{i,d}^+; b_w)\right], \tag{24b}$$

$$\ell_{i,d}^- = p_{i,d} - \underline{w}_d - r, \qquad \ell_{i,d}^+ = \overline{w}_d - r - p_{i,d}. \tag{24c}$$

Here A_o is the response gain, d_o the obstacle interaction range, and b_w the boundary-response buffer. Obstacle directions within the zero-distance guard are omitted. To permit arrival at a separated goal, the summed repulsion is weighted by $\theta_i = \min\{1, d_i/d_g\}^2$, where d_g specifies the goal-approach region. The total acceleration is

$$\mathbf{f}_i = \mathbf{f}_i^{\text{att}} + \theta_i\left(\sum_{j\neq i}\mathbf{f}_{ij} + \mathbf{f}_i^{\text{obs}} + \mathbf{f}_i^{\text{wall}}\right).$$

Coordinate caps $a_{\max}/\sqrt{3}$ and $v_{\max}/\sqrt{3}$ provide a conservative numerical step for the geometric simulation. Clipping this response to the acceleration caps and the next-step velocity caps gives $\mathbf{f}_i^{\text{clip}}$. All UAVs advance synchronously:

$$\mathbf{p}_i^+ = \mathbf{p}_i + \Delta t\,\mathbf{v}_i + \tfrac{1}{2}\Delta t^2\mathbf{f}_i^{\text{clip}}, \qquad \mathbf{v}_i^+ = \mathbf{v}_i + \Delta t\,\mathbf{f}_i^{\text{clip}}. \tag{25}$$

When $d_i < \epsilon_p$ and $\|\mathbf{v}_i\|_2 < \epsilon_v$, the simulated state is set to $(\mathbf{g}_i, \mathbf{0})$ and held there while the remaining UAVs proceed. This gives exact goal knots with a common sequence of time indices. If the cooperative simulation reaches its limit, the static guide paths provide a fallback geometric reference. The same fallback is available to every guided method. A missing guide or an inadmissible fallback time law terminates reference generation. The SFM supplies coordinated geometric references; collision clearance and motion feasibility are established by the subsequent optimization and acceptance stages.

5.2. Common-Horizon Rest-to-Rest Timing

For UAV i, remove repeated consecutive points from its simulated geometric polyline. Let $\mathbf{z}_i(q)$ be its piecewise-linear arc-length parameterization, with $q \in [0, L_i]$. Each nonzero segment has $\|\mathbf{z}_i'(q)\|_2 = 1$. The scalar profile is constructed on the optimization grid:

$$b_k = \min\{v_{\max}, a_{\max} t_k, a_{\max}(T_f - t_k)\}, \qquad Z = \frac{h}{2}\sum_{k=0}^{K-1}(b_k + b_{k+1}), \tag{26a}$$

$$u_{i,k} = \frac{L_i}{Z} b_k, \qquad q_{i,k}^{\star} = \frac{h}{2}\sum_{\ell=0}^{k-1}(u_{i,\ell} + u_{i,\ell+1}), \tag{26b}$$

$$\alpha_{i,k} = (u_{i,k+1} - u_{i,k})/h. \tag{26c}$$

The profile satisfies zero endpoint speeds and covers exactly L_i by T_f. Its maximum speed and absolute acceleration must satisfy $v_{\max}$ and $a_{\max}$. An SFM route that exceeds this capacity triggers the guide fallback; a fallback or straight route that still exceeds it ends the attempt. The horizon is unchanged. The reference states are

$$\widehat{\mathbf{p}}_{i,k} = \mathbf{z}_i(q_{i,k}^{\star}), \qquad \widehat{\mathbf{v}}_{i,k} = u_{i,k}\mathbf{z}_i'(q_{i,k}^{\star}). \tag{27}$$

At a geometric knot, the outgoing tangent is used, with the incoming tangent at the terminal knot. The common timing rule gives each variant the same horizon and boundary-speed conditions. Route-specific arc-length scaling retains the simulated paths but can change their relative passage times; cooperative optimization resolves the remaining temporal conflicts. Bends can also produce vector-dynamics defects. The scalar law establishes reference timing, while final motion feasibility is determined after optimization and reconstruction.

6. Lightweight Iterative Optimization Method

With the flight horizon fixed, LIOM revises the trajectories and their local collision-avoidance regions together. Its corridor reconstruction and external-penalty principles build on [31,32]. The procedure developed here organizes computation into a hard-QP entry and two recovery layers. The first layer relaxes conflicting kinematic and relative-corridor restrictions so that geometry can be updated. The second layer responds to a reconstructed motion that violates a physical limit, using tighter recovery and an exact-kinematics finish. Three outcomes are distinguished throughout: a nonempty relaxed subproblem, numerical recovery of its candidate, and final acceptance of the reconstructed continuous motion.

6.1. Two-Penalty Recovery Formulation

For the double-integrator motion model, the kinematic defect is affine in the position and velocity variables. The two quantities that measure constraint recovery are

$$D = \sum_{i=1}^{N}\sum_{k=0}^{K-1}\|\mathbf{E}_{i,k}\|_2^2, \qquad S = \sum_{(i,j,k)\in\mathcal{A}}\delta_{ij,k}. \tag{28}$$

At iteration ν, the augmented objective is

$$J^{(\nu)} = C + \frac{\rho_q^{(\nu)}}{h} D + \rho_s^{(\nu)} S. \tag{29}$$

The quantities D and S have units of m^2 and m, respectively. The factor $1/h$ accounts for interval length in the kinematic penalty. For a dimensionless augmented objective in

SI coordinates, ρ_q and ρ_s have units $\mathrm{s/m^2}$ and $\mathrm{m^{-1}}$, respectively. Multiplying the entire objective by a positive scalar improves numerical scaling without changing its minimizers. The weights $\rho_q^{(\nu)}$ and $\rho_s^{(\nu)}$ regulate the recovery of kinematics and relative-corridor compatibility, respectively.

Let $\mathcal{I} = \{1, \dots, N\}$, $\mathcal{K} = \{0, \dots, K\}$ and $\mathcal{K}^- = \{0, \dots, K-1\}$. Corridor bounds and $\mathcal{A}^{(\nu)}$ are fixed during an inner solve. The complete subproblem is

$$\min_{\mathbf{P},\mathbf{V},\delta} \quad C(\mathbf{V}) + \frac{\rho_q^{(\nu)}}{h} \sum_{i\in\mathcal{I}} \sum_{k\in\mathcal{K}^-} \|\mathbf{E}_{i,k}\|_2^2 + \rho_s^{(\nu)} \sum_{(i,j,k)\in\mathcal{A}^{(\nu)}} \delta_{ij,k} \tag{30a}$$

$$\text{s.t.} \quad \underline{\mathbf{w}} + r\mathbf{1} \le \mathbf{p}_{i,k} \le \overline{\mathbf{w}} - r\mathbf{1}, \qquad i \in \mathcal{I},\ k \in \mathcal{K}, \tag{30b}$$

$$\underline{\gamma}_{i,k} \le \mathbf{p}_{i,k} \le \overline{\gamma}_{i,k}, \qquad i \in \mathcal{I},\ k \in \mathcal{K}, \tag{30c}$$

$$U\mathbf{v}_{i,k} \le v_{\max}\mathbf{b}, \qquad i \in \mathcal{I},\ k \in \mathcal{K}, \tag{30d}$$

$$U(\mathbf{v}_{i,k+1} - \mathbf{v}_{i,k})/h \le a_{\max}\mathbf{b}, \qquad i \in \mathcal{I},\ k \in \mathcal{K}^-, \tag{30e}$$

$$\mathbf{p}_{i,0} = \mathbf{s}_i, \quad \mathbf{p}_{i,K} = \mathbf{g}_i, \quad \mathbf{v}_{i,0} = \mathbf{v}_{i,K} = \mathbf{0}, \qquad i \in \mathcal{I}, \tag{30f}$$

$$\underline{\boldsymbol{\eta}}_{ij,k} - \delta_{ij,k}\mathbf{1} \le \mathbf{p}_{i,k} - \mathbf{p}_{j,k}, \qquad (i,j,k) \in \mathcal{A}^{(\nu)}, \tag{30g}$$

$$\mathbf{p}_{i,k} - \mathbf{p}_{j,k} \le \overline{\boldsymbol{\eta}}_{ij,k} + \delta_{ij,k}\mathbf{1}, \qquad (i,j,k) \in \mathcal{A}^{(\nu)}, \tag{30h}$$

$$\delta_{ij,k} \ge 0, \qquad (i,j,k) \in \mathcal{A}^{(\nu)}. \tag{30i}$$

In the objective, i ranges over $\mathcal{I}$. Acceleration and $\mathbf{E}_{i,k}$ are substituted expressions. Workspace and STC bounds are intersected, and endpoint states are imposed by equal lower and upper variable bounds. Every constraint is linear or a variable bound. Since C, D are convex quadratic and S is linear, the fixed-corridor program is a convex quadratic program (QP). The affine double-integrator equations and hard relative boxes also retain this convex structure. The two relaxations preserve availability of intermediate feasible points when those hard requirements conflict; corridor reconstruction changes the local geometric region between solves.

Proposition 1 (Nonemptiness of the relaxed fixed-corridor subproblem)**.** *Suppose every spatial box is nonempty, contained in the admissible workspace, and contains the prescribed start or goal at the corresponding endpoint. For the rest-to-rest task, the relaxed subproblem* (30) *has a feasible point with finite objective.*

Proof. Choose any position inside each box, preserve the fixed endpoint positions, and set every velocity to zero. The velocity, acceleration, and rest-to-rest boundary bounds are satisfied. For each active triple, choose

$$\delta_{ij,k} = \max\left\{0,\ \max_d(\underline{\eta}_{ij,k,d} - r_{ij,k,d}),\ \max_d(r_{ij,k,d} - \overline{\eta}_{ij,k,d})\right\}. \tag{31}$$

Both relative inequalities are satisfied. All chosen states and slacks are finite, hence so is the augmented objective. □

Proposition 1 establishes local subproblem nonemptiness under the stated box conditions. The outer loop checks numerical recovery and continuous-motion acceptance separately. A failed construction or an exhausted budget terminates the attempt; a finite candidate can support the next corridor update even when the current solve has not reached numerical recovery.

6.2. Hard-QP Entry and Adaptive Recovery

An initial hard QP provides a fast path when the reference already admits a consistent motion. It retains the spatial boxes and motion bounds, imposes $\mathbf{E}_{i,k} = \mathbf{0}$, and replaces the relative boxes by separating half-spaces

$$\mathbf{n}_{ij,k}^{\top}(\mathbf{p}_{i,k} - \mathbf{p}_{j,k}) \geq R_{ij,k}, \qquad \mathbf{n}_{ij,k} = \frac{\widehat{\mathbf{p}}_{i,k} - \widehat{\mathbf{p}}_{j,k}}{\left\|\widehat{\mathbf{p}}_{i,k} - \widehat{\mathbf{p}}_{j,k}\right\|_2}. \tag{32}$$

Coincident reference nodes use the assigned start separation to define a direction. This entry solve receives a bounded fraction of the planning budget. A candidate that passes reconstruction and complete-interval checks is returned immediately. Otherwise, the two-penalty box formulation supplies the recovery loop. The fallback is triggered by the outcome of the hard solve and its motion check.

For the relaxed loop, normalize the recovery quantities as

$$\bar{D} = \frac{D}{NK}\left(\frac{h_0}{h}\right)^2, \qquad \bar{S} = \frac{S}{\max\{1, |\mathcal{A}|\}}, \qquad \bar{D} \leq \varepsilon_q, \quad \bar{S} \leq \varepsilon_s. \tag{33}$$

Here h_0 is a reference interval length. The defect test measures the mean squared displacement mismatch at a common temporal scale; the slack test measures mean relative-box violation. Their units are m^2 and m. This normalization removes direct summation over team size and active pairs. Final physical acceptance remains a separate, unnormalized test of every returned interval. Each penalty is increased only when its corresponding recovery test fails:

$$\rho_a^{(\nu+1)} = \begin{cases} \min\{\beta_a \rho_a^{(\nu)}, \bar{\rho}_a\}, & Q_a^{(\nu)} > \varepsilon_a, \\ \rho_a^{(\nu)}, & Q_a^{(\nu)} \leq \varepsilon_a, \end{cases} \quad (Q_q, Q_s) = (\bar{D}, \bar{S}). \tag{34}$$

Capping limits the penalty scale while reconstruction changes the local feasible region. A finite bounded candidate seeds the next corridors. If the solver provides no usable iterate, the spatial-box centers provide a shared relocation rule, with fixed endpoints preserved. Static guides remain available to relocate obstacle-intersecting seeds. Hard iterative comparisons receive the same updates and planning budget.

Numerical recovery and a bounded violation of the local linear constraints trigger control reconstruction. If the reconstructed motion fails physical acceptance, the method tightens ε_q, increases the kinematic penalty within its cap, and attempts an exact-kinematics, zero-slack QP in the current boxes. A successful motion is returned; otherwise the relaxed candidate seeds another corridor update. Thus an unsuccessful physical check has an explicit continuation branch. Exhausting the iteration or computation budget returns planning failure.

For a feasible fixed convex problem, a linear slack penalty can be exact above a threshold determined by suitable optimal constraint multipliers. That threshold varies with the current corridors. A quadratic defect penalty generally retains nonzero defects at finite weight. Capped continuation, exact finishing, and complete-interval checks jointly determine the accepted motion. Acceleration effort regularizes each local solution.

6.3. Exact Rest-to-Rest Output from Recovered Controls

Finite kinematic defects are handled directly at the control level. For one spatial coordinate of UAV i, collect candidate accelerations in $\mathbf{a}_i \in \mathbb{R}^K$. In the following two equations,

s_i, g_i, v_i^0, v_i^f denote components along this same coordinate. Constant-acceleration integration expresses terminal velocity and displacement as two control moments:

$$H = \begin{bmatrix} h & h & \cdots & h \\ h^2(K-\frac{1}{2}) & h^2(K-\frac{3}{2}) & \cdots & h^2/2 \end{bmatrix}, \qquad \mathbf{b}_i = \begin{bmatrix} v_i^f - v_i^0 \\ g_i - s_i - T_f v_i^0 \end{bmatrix}. \tag{35}$$

The minimum Euclidean-norm correction satisfying $H\bar{\mathbf{a}}_i = \mathbf{b}_i$ is

$$\bar{\mathbf{a}}_i = \mathbf{a}_i + H^\top (HH^\top)^{-1}(\mathbf{b}_i - H\mathbf{a}_i). \tag{36}$$

The two rows of H are independent for $K \geq 2$, so only a 2×2 linear system is required for each coordinate. The correction requires only a closed-form linear solve.

Starting from the prescribed initial state, the corrected controls generate

$$\bar{\mathbf{v}}_{i,k+1} = \bar{\mathbf{v}}_{i,k} + h\bar{\mathbf{a}}_{i,k}, \tag{37a}$$
$$\bar{\mathbf{p}}_{i,k+1} = \bar{\mathbf{p}}_{i,k} + h\bar{\mathbf{v}}_{i,k} + \tfrac{1}{2}h^2\bar{\mathbf{a}}_{i,k}. \tag{37b}$$

For $0 \leq \tau \leq h$, the returned interval motion is

$$\bar{\mathbf{p}}_i(t_k+\tau) = \bar{\mathbf{p}}_{i,k} + \tau\bar{\mathbf{v}}_{i,k} + \tfrac{1}{2}\tau^2\bar{\mathbf{a}}_{i,k}, \qquad \bar{\mathbf{v}}_i(t_k+\tau) = \bar{\mathbf{v}}_{i,k} + \tau\bar{\mathbf{a}}_{i,k}. \tag{38}$$

It follows immediately from (37) that position and velocity agree at neighboring interval boundaries. Equation (36) enforces both terminal moments, giving the prescribed final position and zero final velocity. The output therefore satisfies (2) on every interval, with continuous velocity and bounded, possibly discontinuous acceleration when the subsequent limit checks pass.

To relate reconstruction to numerical recovery, define

$$D_i = \sum_{k=0}^{K-1} \left\|\mathbf{E}_{i,k}\right\|_2^2, \qquad \mathbf{e}_i = \sum_{k=0}^{K-1} \mathbf{E}_{i,k}, \qquad \|\mathbf{e}_i\|_2 \leq \sqrt{KD_i}. \tag{39}$$

The inequality follows from Cauchy–Schwarz. With exact candidate endpoint bounds and $\mathbf{a}_{i,k} = (\mathbf{v}_{i,k+1} - \mathbf{v}_{i,k})/h$, the uncorrected terminal velocity is exact, and the remaining displacement mismatch is $\mathbf{e}_i$. For any coordinate d, (36) therefore gives

$$\left\|\bar{\mathbf{a}}_{i,d} - \mathbf{a}_{i,d}\right\|_2 \leq \left\| H^\top (HH^\top)^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\|_2 \sqrt{KD_i}. \tag{40}$$

For a fixed grid, the bound vanishes with D_i; the actual correction also depends on cancellation among interval defects. The integrated motion is checked against every obstacle and UAV pair, together with the speed, acceleration, and workspace bounds. These checks account for its displacement from the QP candidate while retaining the prescribed horizon and endpoint states.

6.4. Computational Structure

Acceleration elimination leaves $6N(K+1)$ state components before fixed endpoints are removed. With $M = |\mathcal{A}|$ active pair–node triples, the relaxed model adds M slacks and $6M$ relative inequalities. Fixed-facet motion restrictions contribute $O(NK)$ rows. Sparse objective and constraint assembly therefore requires $O(NK + M)$ storage and work after the active set is known. Pair screening costs $O(N^2K)$ in the worst case. For a fixed expansion limit and step, testing B obstacles during spatial-box construction costs $O(NKB)$, up to the fixed bisection count.

Algorithm 1: LIOM with hard-QP entry and acceptance-driven recovery

Input: Assigned starts and goals, map, T_f, motion limits, and budgets
Output: Accepted joint motion or planning failure

Construct reference geometry and admissible scalar timing
if *SFM generation fails* **then**
    Try static-guide geometry with the same timing rule
if *no admissible reference is available* **then**
    return *planning failure*
Initialize $\rho_q, \rho_s, \varepsilon_q$ and the reference
for $\nu = 0, \ldots, \nu_{\max} - 1$ **do**
    if *budget exhausted* **then**
        break
    Construct spatial boxes, screen distant pairs, and construct relative boxes
    if *boxes are invalid or exclude a fixed endpoint* **then**
        break
    if $\nu = 0$ **then**
        Try the bounded hard-QP entry (32)
        if *its reconstructed motion passes complete-interval checks* **then**
            return *accepted motion*
    Solve the relaxed QP (30)
    if *a finite candidate is available* **then**
        Evaluate $\bar{D}$, $\bar{S}$, and the local linear-constraint violation
        if *numerical recovery and the candidate-readiness test hold* **then**
            Correct endpoint moments and integrate controls
            if *complete-interval checks pass* **then**
                return *accepted motion*
            Tighten ε_q and increase ρ_q within their limits
            Try exact kinematics and zero slack in the current boxes
            if *the resulting reconstructed motion passes all checks* **then**
                return *accepted motion*
    Update the two penalties independently using (34)
    Update the reference from a usable candidate or the spatial-box centers
return *planning failure*

The complete-interval checker considers all pairs and obstacles, with a fixed-degree polynomial calculation per interval, giving $O((N^2 + NB)K)$ work. QP solution time also depends on sparse factorization, numerical conditioning, and the required iterations. Geometric screening reduces actual coupling, but the centralized worst-case dependence remains quadratic in team size. Section 7.6 measures both team and grid scaling.

7. Simulation Setup, Results, and Discussion

7.1. Tasks and Evaluation Protocol

The evaluation preserves the fixed completion time, assigned endpoints, footprint, and physical limits across methods. The original 30 tasks comprise crossing, passage, and obstacle-field families, with five difficulty levels and two layouts per level. Crossing teams grow from three to seven UAVs; the other families use five UAVs. The square passage opening decreases from 4.4 to 2.8 m, and the four-obstacle size multiplier increases from 0.50 to 1.10. The workspace is $[-15, 15]^2 \times [0.5, 7.5]$ m, and every main-benchmark flight lasts 12 s.

Additional physical tasks use a smaller $[-8, 8]^2 \times [0.5, 7.5]$ m workspace. Crossing teams contain 10, 15, or 20 UAVs, with ten randomized start–goal realizations at each size. Passage and obstacle-field tasks use ten UAVs, three geometric levels, and five realizations per level. The passage openings are 3.2, 2.6, and 2.0 m; obstacle dimensions

increase across levels. These 60 tasks change physical congestion while retaining the nominal corridor-growth limits. A separate targeted passage study considers teams of five, eight, and ten UAVs at a 16 s horizon. The accompanying data provide the task definitions and outcomes.

The principal methods are **LIOM (proposed)**, with a hard-QP entry and the complete recovery loop; **Soft LIOM**, which starts directly with the relaxed loop; **Hard box–iterative**, which imposes exact kinematics and zero relative-box slack while retaining reconstruction; **Hard RSC–iterative**, which uses exact kinematics and separating half-spaces with the same reconstruction opportunities; and **Hard box–once**, which solves only the initial hard-box QP. One-shot RSC and one-shot soft variants provide further ablations. All share the reference policy, spatial construction, safe pair screening, physical acceptance, and total budget. These task-matched hard and soft corridor variants examine the iterative mechanisms associated with [21,31,32] in the same UAV model.

The archived comparisons use a 1 s hard-entry allowance. Section 7.7 evaluates 1, 3, and 5 s allowances and establishes the recommended 3 s setting for new runs. The main comparison has three repetitions per scene and method, with rotated method order; additional one-shot variants have one repetition. The repetitions characterize runtime variability. Paired case records identify whether a method returns an accepted motion within the budget. CPU time includes MATLAB and the OSQP worker; wall time includes reference generation, construction, numerical assembly, solver setup and exchange, solution, reconstruction, and verification. Process startup and plotting are excluded. All attempts, including failures, enter runtime summaries. The 30 s budget is a nominal stopping rule checked between stages. The results distinguish accepted motions returned within 30 s, accepted motions returned later, and attempts without an accepted motion.

7.2. Physical, Numerical, and Implementation Settings

The UAV is a generic hovering-capable multirotor with a spherical planning radius of 0.25 m and required surface clearance of 0.10 m. The translational limits are 4 m/s and 3 m/s^2. These values define a moderate flight envelope for the planning and tracking studies. The main grid has $K = 60$ intervals. Table 2 gives the common numerical settings.

Table 2. Common settings. Linear-row tolerances use the corresponding native SI units. Final acceptance retains the physical tolerances below.

Parameter	Value	Parameter	Value
T_f / K	12 s / 60	w_a	0.05
r / c	0.25 / 0.10 m	$v_{\max}$ / $a_{\max}$	4 m/s / 3 m/s^2
ℓ_s / ℓ_r	1.5 / 1.2 m	Expansion increment	0.15 m
$\rho_q^{(0)}$, $\rho_s^{(0)}$	10^3	Penalty caps / growth	10^5 / 2
ε_q / h_0	10^{-3}/300 m^2 / 0.2 s	ε_s	10^{-4} m
Candidate row tolerance	10^{-3}	OSQP absolute / relative tolerance	10^{-6} / 10^{-6}
Maximum outer rounds	12	Hard entry / regular solve	1 / 5 s (archived runs)
Nominal planning budget	30 s	Geometric guard	10^{-5} m
Clearance tolerance	2×10^{-5} m	Speed tolerance	2×10^{-5} m/s
Acceleration tolerance	2×10^{-5} m/s^2	Endpoint / integration tolerance	10^{-9} m, 10^{-9} m/s

Sparse QPs are assembled numerically in MATLAB R2021b and solved by OSQP [36]. The computer uses an Intel Core i9-12900H processor and 16 GB RAM. OSQP checks candidate readiness in blocks of 250 iterations, using the local linear-row tolerance and, for

relaxed candidates, both recovery thresholds. Readiness permits reconstruction; complete-interval physical checking determines acceptance. An identical-matrix OSQP audit compares candidate-readiness stopping with ordinary solver termination at two numerical tolerances.

The initial penalty scale favors consistency while the cap limits conditioning problems. The defect threshold uses a five-UAV, 60-interval reference scale; its normalized form permits team and grid comparisons. The control-effort coefficient sets its relative importance in the augmented objective. Grouped sensitivity tests vary these choices around the development settings. The static guide grid is 0.65 m. SFM uses a 0.02 s integration step, a 2 m/s preferred speed, gains $A_p = A_o = 4$ m/s^2, decay length 0.35 m, look-ahead 0.35 m, response time 0.3 s, and tangential factor 0.5. All remaining settings and the frozen scenario definitions accompany the experiment scripts.

The SFM samples form geometric polylines. Repeated points are removed, and arc-length interpolation evaluates (27) on the 0.2 s optimization grid. The prescribed scalar law therefore determines geometric resampling onto the optimization grid. Initialization time includes SFM, guide construction, timing, and any guide fallback.

7.3. Accepted Motion and Main-Benchmark Performance

Complete-interval checks are applied to the integrated motion (38). Squared pair distance is quartic in interval time, so its extrema follow from a cubic derivative. Point-to-AABB distance is minimized over box-face crossing intervals and their stationary points. Workspace extrema, endpoints, and integration identities are checked as well. Speed is affine within each interval and acceleration is constant, allowing exact interval checks of their Euclidean limits. These tests examine every pair and obstacle, including pairs screened out of the QP.

Table 3 reports computation over the main benchmark. Proposed LIOM has mean wall time 7.83 s, compared with 4.03 s for Hard RSC–iterative. The bounded hard entry can route an attempt to recovery before a longer hard solve would finish, so its runtime includes the cost of maintaining that option. The following winding tasks identify the specific local incompatibilities for which the additional route is useful. All task-level outcomes are retained with the timing records.

Table 3. Computation on the 30-scene main benchmark with the original 1 s LIOM hard entry and three executions per method and scene. Means and the wall-time P95 include all 90 attempts, including budget stops. Rounds and QPs are mean counts.

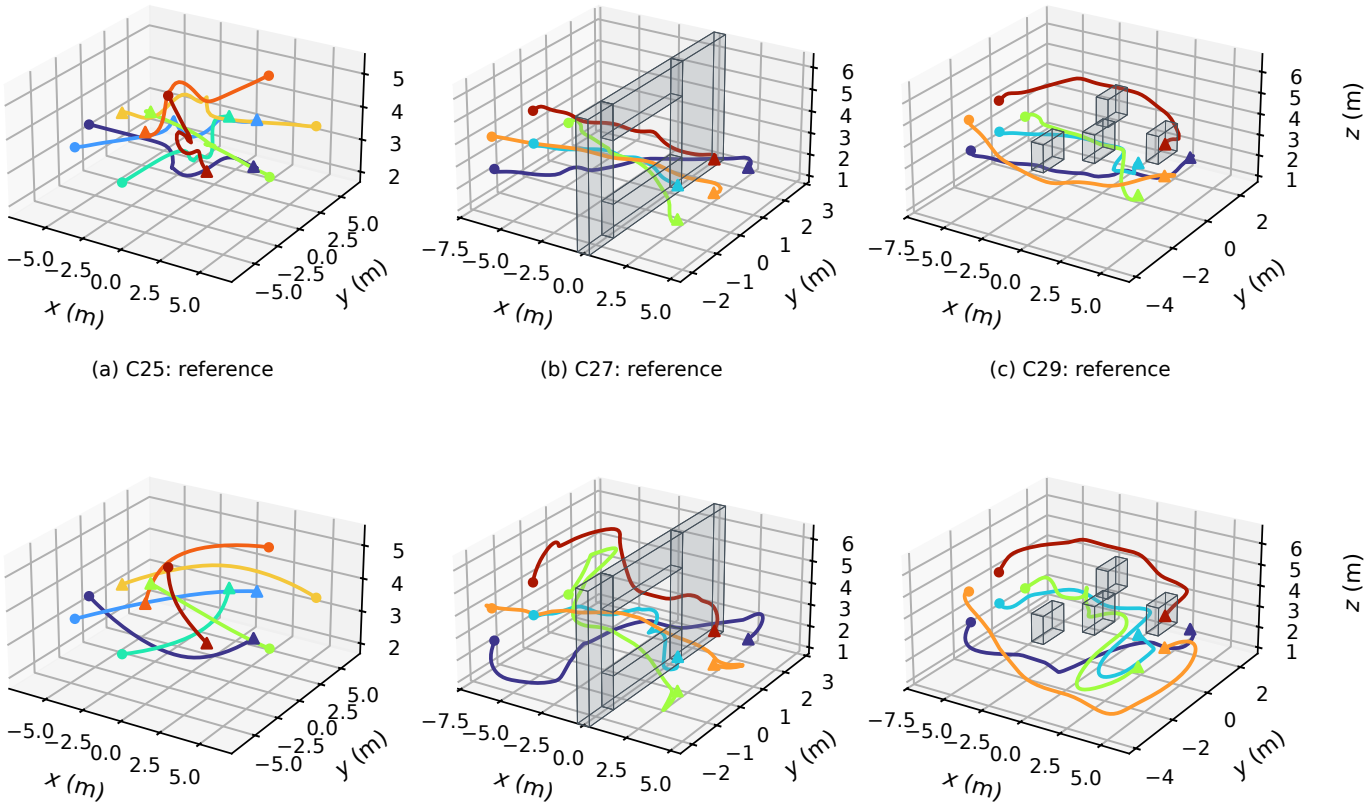
Method	CPU (s)	Wall (s)	P95 (s)	Rounds	QPs
LIOM (proposed)	8.66	7.83	25.39	2.08	2.46
Soft LIOM	8.84	8.42	26.92	2.27	2.29
Hard box–iterative	16.08	14.45	30.06	3.92	3.92
Hard RSC–iterative	4.56	4.03	10.56	1.44	1.44
Hard box–once	5.37	5.29	13.57	1.00	1.00

7.4. Recovery in Winding Counterflow Tasks

Three full-height walls with alternating openings produce repeated turns and opposing traffic. The representative task contains five UAVs, 2.2 m openings, and a fixed 24 s flight horizon with $h = 0.2$ s. The spatial and relative growth limits, motion envelope, footprint, acceptance thresholds, and 30 s planning budget remain unchanged. A common 0.325 m static-guide grid resolves the openings. Each method receives the same saved reference and is charged the same measured initialization time. Hard RSC–iterative retains its reconstruction loop.

Table 4. Accepted proposed-method motions in the first main-benchmark repetition (29 scenes). Integration errors quantify numerical consistency. Clearance and motion limits are checked separately over complete intervals.

Quantity	Observed extremum
Minimum inter-UAV center distance (m)	1.388
Minimum obstacle surface clearance (m)	0.4959
Maximum speed (m/s)	3.325
Maximum acceleration (m/s^2)	3
Maximum endpoint coordinate error (m)	1.199e-14
Maximum integration-identity error	8.882e-16
Maximum control correction (m/s^2)	9.663e-05
Maximum coordinate displacement from QP nodes (m)	0.01585



(a) C25: reference (b) C27: reference (c) C29: reference

(d) C25: accepted motion (e) C27: accepted motion (f) C29: accepted motion

Figure 3. Representative accepted cooperative motions with metric axes. Starts are circles, fixed goals are triangles, and colors identify UAVs. The full 30-scene overview is provided separately in the reproducibility package.

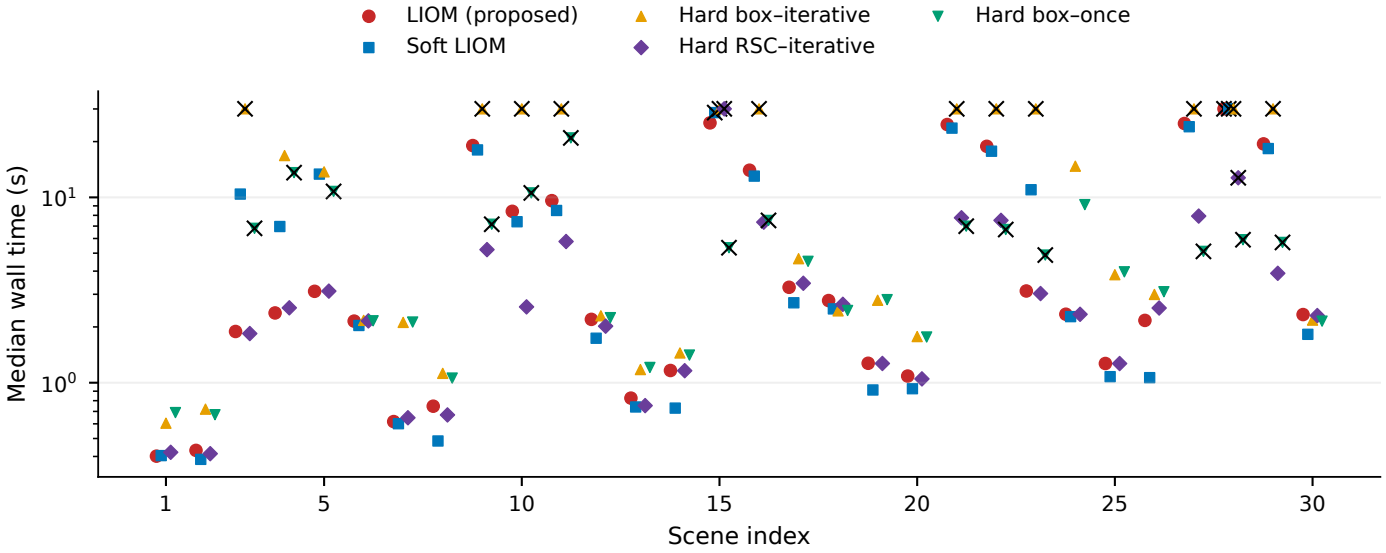


Figure 4. Main-benchmark computation. Per-scene median wall times use separated method positions and a logarithmic axis. Crosses mark a scene–method combination with an incomplete attempt among its three repetitions. All attempts enter the timing summaries.

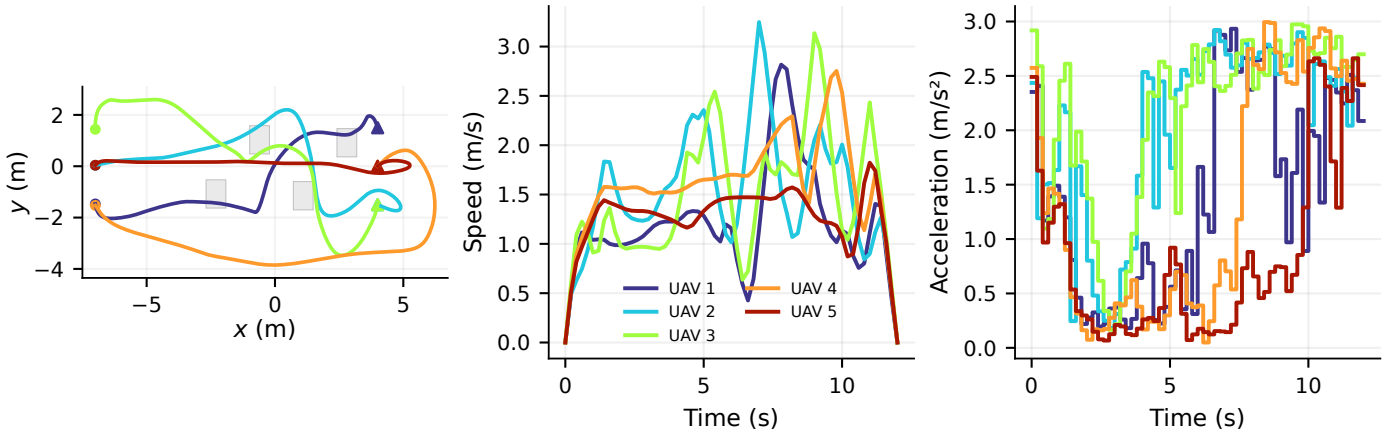


Figure 5. Planar projection and individual UAV speed and acceleration histories for a representative accepted motion. Colors identify UAVs consistently across the panels.

An exploratory search identified the task in Figure 6. In the three initial paired executions of this task, proposed LIOM accepted the motion after five recovery rounds in 26.59–27.18 s. Soft LIOM also accepted all three executions, taking 25.33–25.72 s. Hard RSC–iterative completed eight reconstruction rounds per execution without acceptance within the budget. These repetitions characterize one physical task. Section 7.5 reports a further paired execution alongside the nonlinear and sequential-convex methods. The accepted motion has minimum center separation 1.086 m, minimum obstacle surface clearance 0.495 m, maximum speed 3.577 m/s, and maximum acceleration 2.990 m/s^2; the respective requirements are 0.60 m, 0.10 m, 4 m/s, and 3 m/s^2.

An independent phase-I linear program minimizes a common row-scaled violation of the initial hard-RSC constraints. Its primal and dual objectives both equal 0.21915, with primal residual below 5×10^{-14} and dual stationarity residual below 2×10^{-17}. Removing the relative rows alone, or the kinematic equalities alone, gives zero phase-I objective. These diagnostic checks identify an incompatibility between the selected relative restrictions and exact motion consistency. The complete returned motion satisfies both requirements after reconstruction. The diagnostic relaxations are used only to identify the conflict.

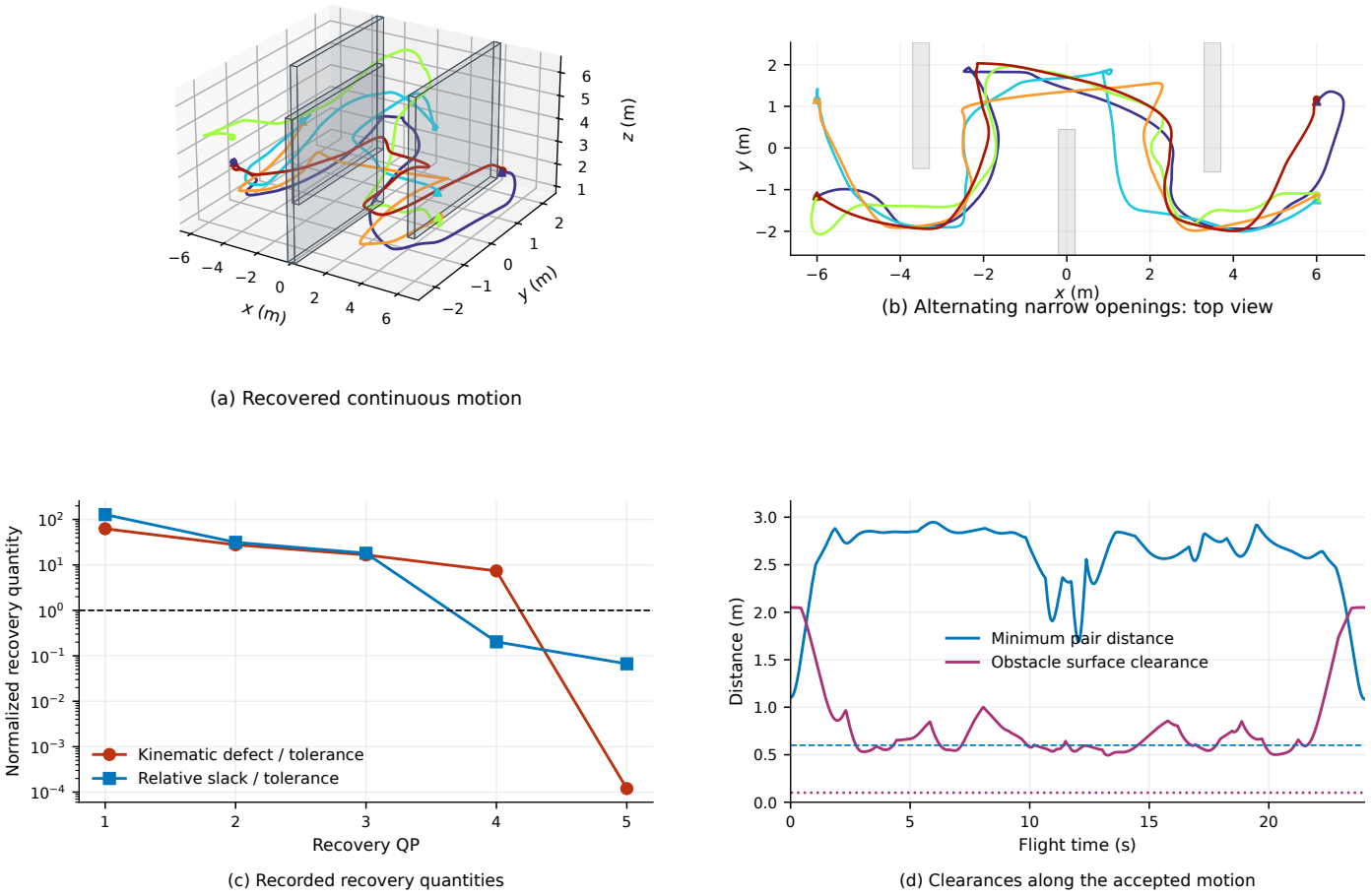


Figure 6. Recovery in a winding counterflow task. Colors identify UAVs. The recovery quantities are divided by their acceptance thresholds; the dashed unit line marks numerical recovery. The clearance curves visualize the integrated motion, whose acceptance is determined by complete-interval checks.

A separate set of 12 previously untested random layouts contains a second recovered task whose initial hard QP has matching phase-I primal and dual objectives of 0.35420. The proposed method returns a checked motion for this task, while iterative hard RSC exhausts the common budget. This independently generated case supports the same conflict mechanism. The complete 12-layout outcomes and all exploratory records accompany the data.

Figure 7 varies physical opening width, team size, and prescribed flight time around the identified geometry. The nominal corridor-growth limits are unchanged. Both methods accept the three-UAV tasks. With five UAVs and 2.2 m openings, proposed LIOM additionally accepts the 24 s and 28 s tasks. Widening the openings permits both methods to succeed. The map locates the additional recovery region under physical congestion; it also shows where neither method returns an accepted motion within the common budget.

7.5. Task-Matched Nonlinear and Sequential Convex Comparisons

Three additional strategies are adapted to the same UAV task. ASCO-3D [28] changes the active collision-risk range, retaining the preceding successful reference after a failed reduced solve. Direct NLP imposes all sampled collision restrictions simultaneously. SCP-3D [30] uses the distance-norm tangent restrictions of Eq. (17) in [30], with supporting

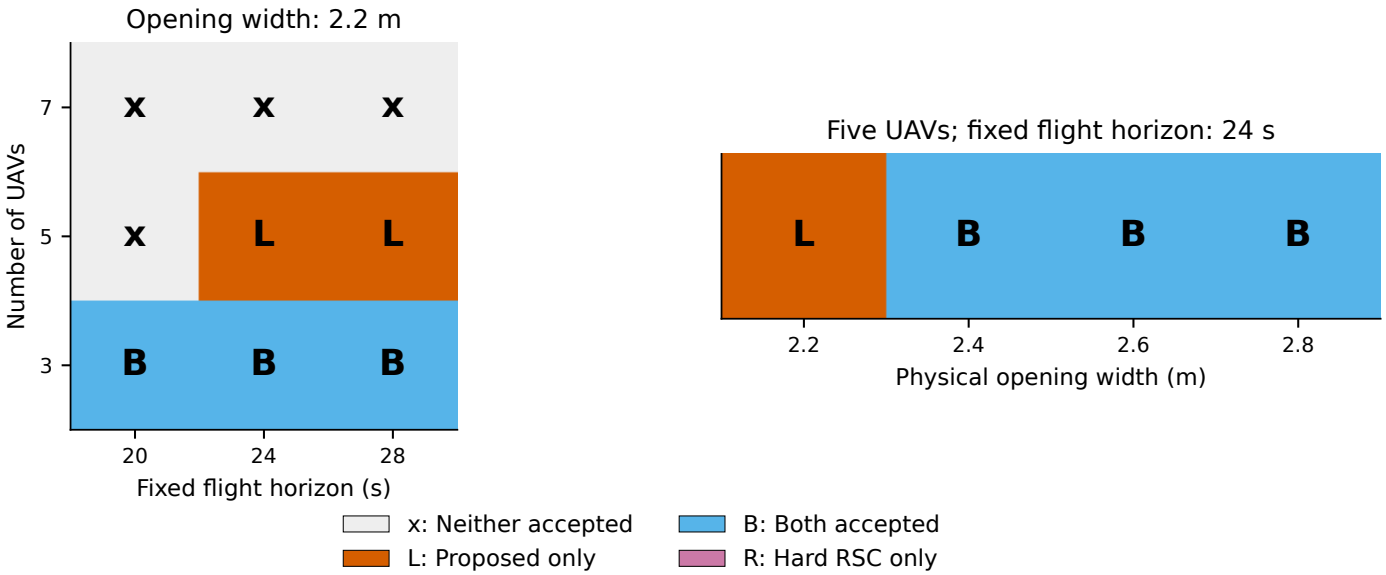


Figure 7. Exploratory physical-parameter map around the same winding layout. Letters supplement color: B, both accepted; L, proposed LIOM only; R, Hard RSC–iterative only; x, neither accepted. Twelve distinct settings are evaluated; the five-UAV, 24 s, 2.2 m setting appears in both panels. Every setting uses the same reference for both methods and the same 30 s budget.

planes for the static boxes. All use hard integration equations, the same polyhedral motion limits, fixed endpoints, fixed horizon, and control-effort objective. Collision restrictions are evaluated at quarter-interval arc samples with reachable-motion padding. The returned motion undergoes the same endpoint reconstruction and complete-interval acceptance tests as LIOM.

The nonlinear ASCO and direct-NLP instances use CasADi [37] and Ipopt [38] with the ma57 linear solver. Every quadratic subproblem uses OSQP. SCP assembles its sparse matrices directly; ASCO iterations without active nonlinear rows also use the QP backend. ASCO's original risk-range lengths are scaled by the ratio of UAV radius to the vehicle-disc radius in the source implementation. Its update rule and 50-round ceiling are retained; the common 30 s budget governs the comparison.

The paired study comprises six original benchmark tasks, both winding layouts, and 15- and 20-UAV tasks. The winding tasks use $T_f = 24$ s and $K = 120$; the remaining tasks use $T_f = 12$ s and $K = 60$. The methods are evaluated serially. Each method receives the identical saved reference and is charged its measured generation time. The comparison records one execution per task–method pair. The nonlinear baselines are rerun with ma57; the unchanged QP implementations retain their recorded measurements. Construction, solution, and final checking all enter the reported time, including calls that overrun the nominal stopping budget.

Table 5 reports the ma57 comparison. LIOM completes C01, C03, C05, and C25 in 1.29–3.14 s. In C27, LIOM returns an accepted motion in 24.35 s, while ASCO, hard RSC, and SCP complete the same task in 8.61, 7.55, and 5.41 s. The ma57 direct-NLP solve also accepts C27, in 27.79 s. These results show how the direct hard path and the recovery path distribute computation across tasks.

Winding A combines three alternating openings with opposing traffic. LIOM and iterative hard RSC return accepted motions in 26.36 and 25.14 s in this comparison. ASCO's candidate motions continue to fail collision checks when its budget is exhausted; SCP reports infeasibility for its initial tangent problem. The hard-RSC result supplements the earlier repetitions in Section 7.4. The positive phase-I certificate identifies an incompatible initial local problem, and later reconstruction can reach a feasible corridor configuration.

Table 5. Task-matched planning times in seconds, including recorded initialization. LIOM uses the original 1 s hard entry in this comparison; the nonlinear columns are rerun with Ipopt/ma57. A cross denotes no accepted motion; † denotes an accepted motion returned after the nominal 30 s budget. Each entry is one recorded execution.

Task	N	**LIOM (1 s)**	Hard RSC	ASCO-3D	Direct NLP	SCP-3D
C01	3	**1.72**	0.90	5.25	2.61	1.77
C03	5	**1.51**	1.42	3.50	29.05	3.27
C05	5	**3.14**	3.38	16.33	33.58†	5.47
C25	7	**1.29**	1.23	21.35	9.33	12.52
C27	5	**24.35**	7.55	8.61	27.79	5.41
C29	5	**19.54**	4.07	15.06	27.16	10.32
Winding A	5	**26.36**	25.14	× 33.72	× 47.10	× 28.41
Winding B	5	**31.92**†	× 30.07	× 31.50	× 31.36	× 30.05
15-UAV	15	**27.49**	29.97	× 34.07	59.34†	19.91
20-UAV	20	**30.12**†	27.70	× 31.27	× 106.75	× 30.06

In Winding B, LIOM produces a physically accepted motion at 31.92 s; the ma57 baselines and the other hard comparisons return no accepted motion in their recorded attempts. The late LIOM output is marked explicitly in the table.

For the 15-UAV task, LIOM, hard RSC, and SCP complete in 27.49, 29.97, and 19.91 s. Direct NLP returns an accepted motion after its nominal budget, and ASCO ends without an accepted motion. At 20 UAVs, hard RSC completes in 27.70 s and LIOM returns an accepted motion at 30.12 s. Symbolic construction consumes the remaining budget in the two winding direct-NLP tasks and the 20-UAV direct-NLP task. The source records separate construction, numerical solution, and physical checking.

7.6. Physical Congestion and Computational Scaling

The physical-congestion study varies crossing-team size, opening width, and obstacle dimensions while retaining the nominal corridor-growth limits. Figure 8 reports the mean computation in each group, including attempts that reach the stopping budget. The 30 s plateau marks termination at the nominal computational budget. These measurements characterize computational load, while the paired winding tasks provide direct evidence of recovery from incompatible local constraints.

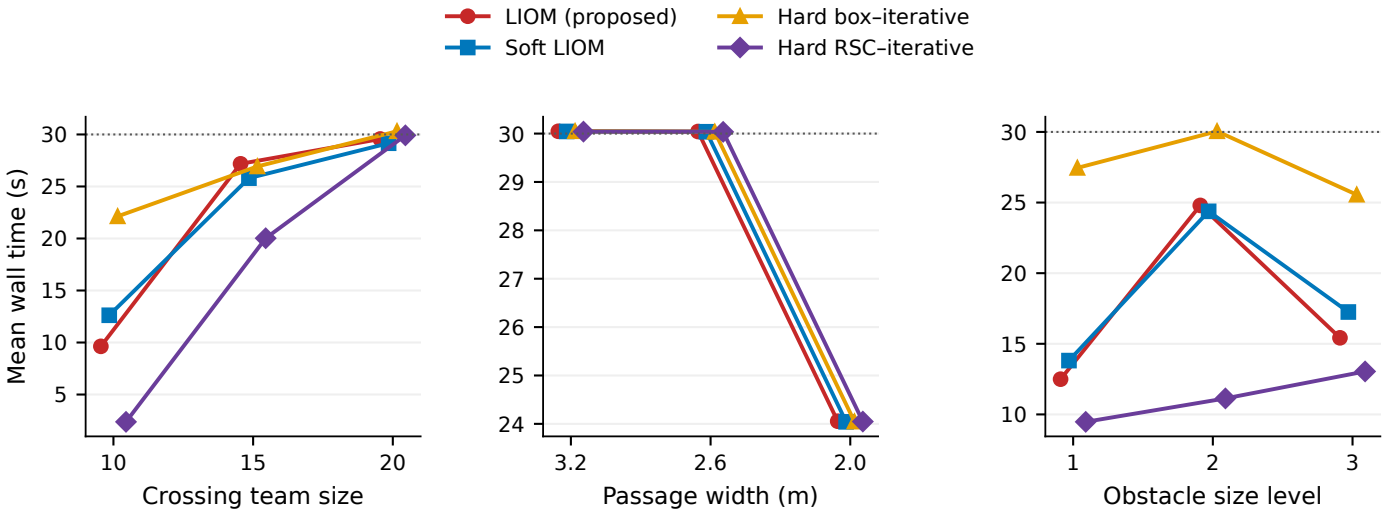


Figure 8. Computation under physical congestion with fixed corridor-growth settings. Panels vary crossing-team size, passage width, and obstacle size. Symbols distinguish methods in grayscale, and all attempts enter each mean.

The grid study combines $N \in \{5, 10, 15, 20\}$ and $K \in \{30, 60, 90\}$ with two predetermined realizations per combination. This separates team growth from discretization changes. Figure 9 relates measured computation to problem size and shows the stage breakdown. The number of pair restrictions can decrease substantially after spatial-box screening, while complete-interval validation still checks all pairs. The reduced-horizon

Table 6. Team/grid scaling over two realizations per setting. Mean wall times include all attempts. Variables and rows are the proposed method's mean maximum dimensions; coupling varies with the spatial boxes.

N	K	LIOM wall (s)	Hard RSC wall (s)	Variables	Rows
5	30	0.16	0.13	870	7,710
5	60	0.21	0.18	1,770	15,497
5	90	0.46	0.44	2,670	23,285
10	30	21.82	9.92	2,653	19,641
10	60	1.01	1.01	3,540	32,012
10	90	11.13	3.66	7,759	57,477
15	30	30.05	25.00	4,770	34,200
15	60	22.38	5.95	9,249	66,477
15	90	25.88	27.80	13,698	98,568
20	30	30.04	30.02	7,476	52,299
20	60	26.98	21.87	14,379	100,914
20	90	21.75	30.06	21,233	149,238

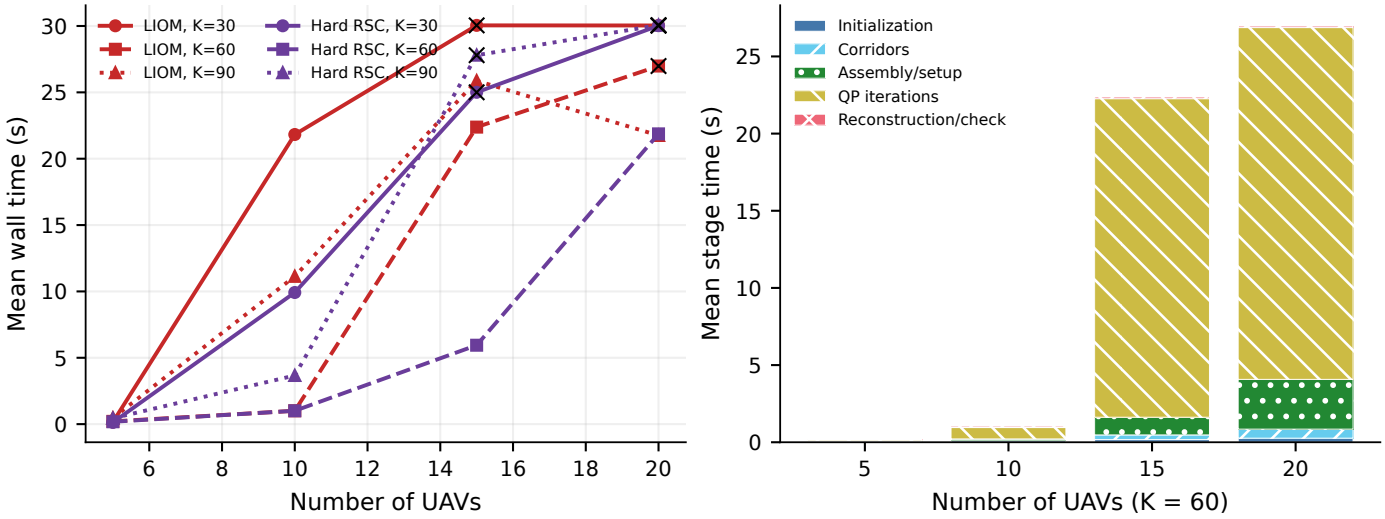


Figure 9. Team/grid scaling and planning-time components. Crosses mark team–grid groups with at least one attempt not accepted within 30 s. The stage breakdown includes initialization, corridor construction, assembly and solver setup, QP iterations, and final checking.

study varies T_f from 8 to 12 s on six fixed geometries with unchanged corridor-growth caps. The accompanying records contain the candidate checks, outer rounds, and stopping outcomes for every method and setting.

7.7. Mechanism and Parameter Studies

A separate entry-budget study varies the initial hard-QP allowance between 1, 3, and 5 s on the ten tasks used in Table 5. All other settings and the total planning budget remain fixed. Table 7 reports the complete runs, including recovery after the entry solve. In C29, the 3 s entry returns directly and reduces total time from 19.27 to 3.63 s and QP calls from five to one. The 3 s setting also returns accepted 15- and 20-UAV motions within budget in this study. Giving the initial hard problem 5 s reduces the time available for subsequent recovery: Winding A then returns its accepted motion after the total budget. The 3 s

allowance offers a useful compromise and is recommended for new runs. The earlier benchmark and scaling records retain their original 1 s allowance; their recorded times are unchanged. The hard-iterative comparison holds reconstruction opportunities fixed. The

Table 7. Hard-entry budget ablation. Each cell gives total planning time in seconds and QP calls. All three settings share the recorded reference and a 30 s total budget, with rotated execution order. × marks no accepted motion and † marks late acceptance.

Task	1 s entry: time / QPs	**3 s entry: time / QPs**	5 s entry: time / QPs
C01	1.75 / 1	0.95 / 1	0.88 / 1
C03	1.35 / 1	1.44 / 1	1.46 / 1
C05	2.84 / 1	2.91 / 1	2.89 / 1
C25	1.20 / 1	1.18 / 1	1.17 / 1
C27	24.08 / 6	26.15 / 6	28.29 / 6
C29	19.27 / 5	3.63 / 1	3.65 / 1
Winding A	28.70 / 5	30.00 / 5	$31.33^{\dagger}$ / 5
Winding B	$30.48^{\dagger}$ / 6	× 30.07 / 5	× 30.04 / 5
15-UAV	× 30.08 / 7	29.09 / 6	× 30.66 / 6
20-UAV	28.37 / 6	27.04 / 6	$30.35^{\dagger}$ / 6

Table 8. Soft-loop acceptance feedback and motion-envelope studies on six scenes. Feedback combines tolerance tightening with an exact-kinematics finish; its components are evaluated jointly.

Variant	Wall (s)	QP calls
Aggregate, no feedback	1.12	1.33
Aggregate + feedback	1.12	1.33
Normalized, no feedback	1.07	1.17
Normalized + feedback	1.07	1.17
cube motion envelope	1.03	1.33
hull14 motion envelope	1.08	1.17
hull26 motion envelope	1.16	1.17

hard-entry ablation compares proposed LIOM with Soft LIOM. The acceptance-feedback study examines how rejected reconstructed motions affect subsequent computation; it also records rejection reasons and additional exact-QP calls.

The six anchor tasks are accepted under all four recovery-criterion/feedback variants. Across the eight frozen suites, 23 reconstructed soft candidates fail a physical check, in 19 attempts; 15 of those attempts later return an accepted motion within budget. The exact-finishing branch returns 9 accepted motions. Across the 1,212 attempts in the eight frozen suites, four initialization failures concern one physical scene evaluated by four methods. The static-guide fallback follows 93 SFM simulation timeouts; no scalar-timing failure is recorded in these suites. Counts refer to attempted executions, including timing repetitions. The initializer study crosses SFM, straight geometry, and static-guide geometry

Table 9. Reference geometry and later guide availability crossed on six anchor scenes. Mean times and rounds include all attempts; task-level outcomes accompany the data.

Reference	Relocation guide	Wall (s)	Rounds
SFM + guide fallback	False	1.09	1.17
SFM + guide fallback	True	1.08	1.17
Straight	False	18.76	9.00
Straight	True	17.98	7.17
Static guide	False	17.95	7.17
Static guide	True	17.93	7.17

with guide availability during later relocation. It separates reference construction from the use of a guide after the initial solve. Penalty, tolerance, corridor-size, and SFM sensitivity

Table 10. One-factor sensitivity on six anchor scenes, all accepted at the tested settings. Columns give mean wall time after halving or doubling one parameter; the last row uses the stated growth factors. Nominal time is 1.07 s.

Changed factor	Lower value: wall (s)	Higher value: wall (s)
Control effort w_a	1.00	1.08
Initial defect penalty	1.08	1.01
Initial slack penalty	1.07	1.08
Defect penalty cap	1.08	1.08
Slack penalty cap	1.07	1.08
Normalized defect tolerance	1.07	1.07
Mean slack tolerance	1.07	1.07
Spatial growth cap	1.06	1.10
Relative growth cap	2.51	1.09
SFM repulsion	1.08	1.09
SFM decay length	1.14	3.60
SFM look-ahead	6.06	1.08
SFM response time	1.10	1.08
SFM tangential fraction	1.16	1.07
Candidate row tolerance	1.08	1.08
Penalty growth (1.5 / 5)	1.08	1.08

Table 11. OSQP audit on identical matrices from three anchor scenes. Readiness stopping is compared with ordinary termination at two tolerances. The feasible-column criterion is maximum linear-row violation $\leq 10^{-3}$; full trajectories still require physical acceptance.

QP	Termination	Row test	Max. violation	Time (s)	Iterations
soft	Readiness	3/3	5.2e-05	1.94	6,833
soft	Standard (10^{-6})	3/3	5.2e-05	1.96	6,833
soft	Standard (10^{-7})	3/3	1.4e-05	6.40	20,000
hard	Readiness	2/3	1.5e+10	1.64	5,808
hard	Standard (10^{-6})	2/3	1.5e+10	4.10	13,767
hard	Standard (10^{-7})	2/3	1.5e+10	4.93	16,308
rsc	Readiness	3/3	8.7e-04	0.45	1,400
rsc	Standard (10^{-6})	3/3	1.6e-15	1.37	4,733
rsc	Standard (10^{-7})	3/3	1.6e-15	2.80	9,283

tests change one factor at a time on six anchor tasks. Corridor-size changes are treated as a numerical sensitivity study; the physical-congestion comparisons retain the nominal limits. The motion-envelope study compares the original cube with 14- and 26-vertex inner approximations. The acceptance study compares aggregate and normalized recovery criteria, with and without acceptance feedback.

7.8. Rigid-Body Tracking in a Physics Simulator

After centralized planning, each UAV receives its time-indexed position, velocity, and acceleration reference. A shared execution clock preserves their temporal coordination. Individual feedback controllers track these references; all inter-UAV decisions have already been coupled in the centralized solve. This architecture assumes known initial states and a common static map.

MuJoCo [39] provides six-degree-of-freedom rigid-body dynamics, gravity, bounded rotor actuation, motor lag, and contact detection. The generic quadrotor has mass 1 kg, inertia $\mathrm{diag}(0.006, 0.006, 0.010)$ kg m^2, rotor coordinates $(\pm 0.14, \pm 0.14, 0)$ m, and a 5 N force limit per rotor. Motor lag is 0.02 s and the integration step is 0.002 s. Position/velocity feedback and a rotation-matrix attitude controller generate four rotor-force commands. The perturbed condition increases mass by 10%, motor lag by 50%, and adds sinusoidal

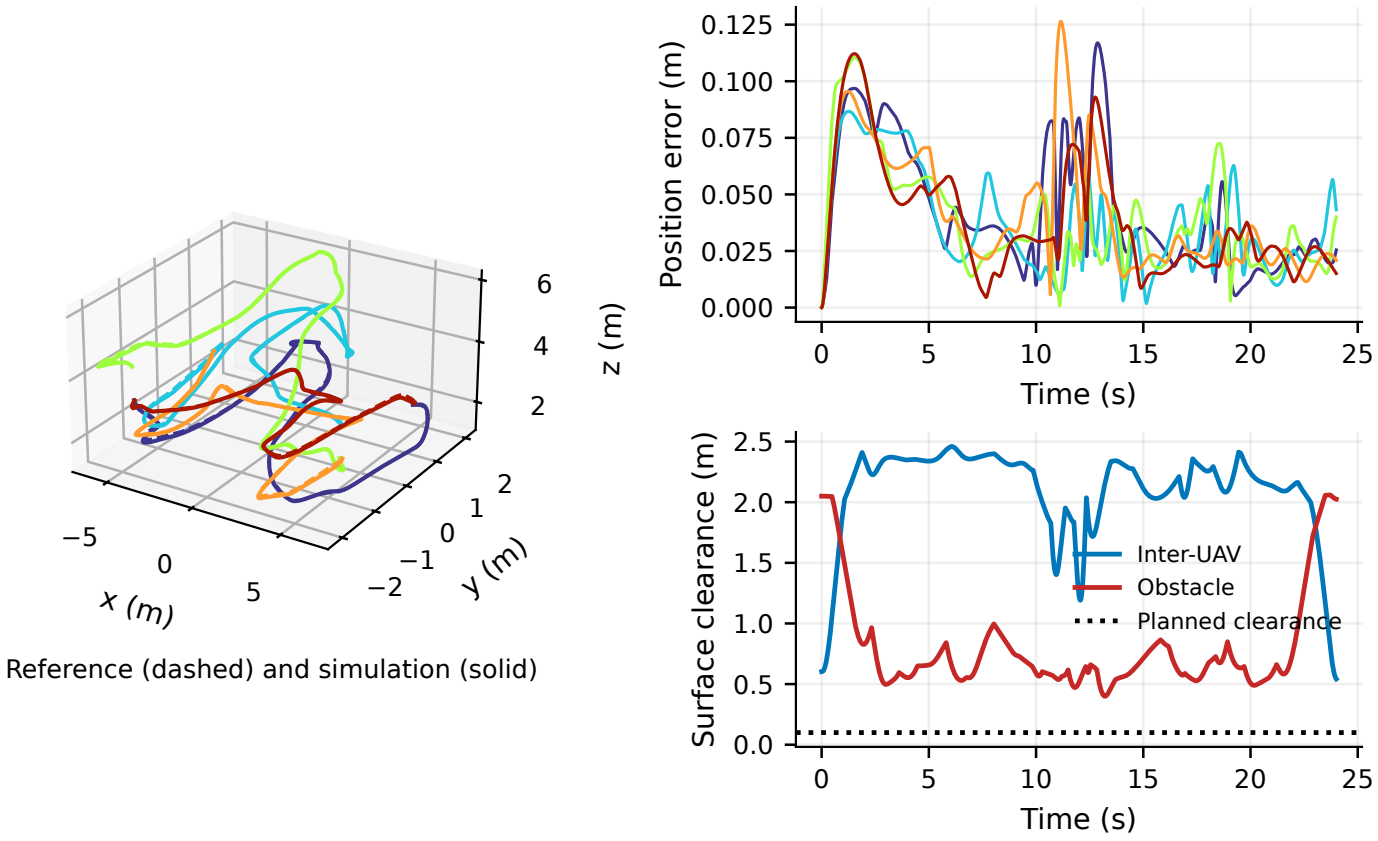


Figure 10. Rigid-body execution of a planned cooperative motion: reference and simulated positions, tracking error, and surface clearance. Simulation measurements are sampled at the physics integration rate.

wind force with 0.15 N amplitude per coordinate. Controller settings are fixed across the evaluated scenes.

The simulator evaluates all accepted proposed-method motions from the first main-benchmark repetition and predetermined accepted larger-team and winding-recovery examples. The same controller is used in the nominal and perturbed conditions. Table 12 reports position error, terminal accuracy at the specified completion time, separation, and contact outcomes. Full time histories, simulator models, and settings accompany the results.

Table 12. MuJoCo tracking over the prescribed flight interval. Terminal error is measured at T_f, before the separately logged settling interval. Clearance is center distance minus two planning radii.

Condition	Motions	Max. error (m)	Mean RMS (m)	Terminal (m)	Pair gap (m)	Contact runs
Nominal	35	0.132	0.017	0.105	0.504	0
Perturbed	35	0.139	0.057	0.121	0.469	0

The closed-loop simulations assess how the planned motions are tracked under rotor-force limits, motor lag, and disturbances. Tracking errors and surface clearances characterize execution by the feedback controller. The planning model and continuous-interval checks supply the corresponding reference-motion assessment.

8. Conclusion

This paper develops a layered solution procedure for fixed-horizon multi-UAV cooperative flight. A hard relative-corridor QP handles directly compatible local problems. Separate kinematic and relative-corridor penalties support further geometric updates when that path does not return an accepted motion. Acceptance-driven tolerance adjustment and exact finishing provide a second recovery layer. Endpoint moment correction, exact integration, and complete-interval checks produce the returned rest-to-rest motions.

The winding counterflow study identifies incompatible initial hard corridors and records accepted motions obtained by relaxed reconstruction at the same flight horizon. Independent phase-I checks confirm incompatibility of the initial hard constraints in two recovered layouts; the accepted continuous motions then establish feasibility of those physical tasks. The broader experiment records also contain nine accepted exact-finishing motions after soft-candidate checks fail. These observations demonstrate the role of both recovery layers. Hard relative corridors remain faster on ordinary compatible cases, motivating their direct entry into the planning procedure.

Team and grid studies characterize computational growth through 20 UAVs. Rigid-body simulations connect the accepted plans to bounded rotor actuation and feedback tracking under disturbances. Future work will extend the known-static-map setting to rotor downwash, uncertain perception, and flight experiments.

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